SentryScan: Assignment 4.2 — Knowledge Representation & API Enrichment

Course: 21CSE525T — Agent Technology
Assignment: 4.2 — Knowledge Representation & API Enrichment
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Repository: https://github.com/drgeekg/SentryScan
Target Branch: `assignment/4.2-knowledge-enrichment`
Git Tag: `assignment-4.2`
Status: Completed & Verified

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1. Executive Summary & Theoretical Context

Assignment 4.2 integrates formal **Knowledge Representation & External API Enrichment** (Russell & Norvig, Ch. 12) into SentryScan, allowing the agent to augment sparse perceptual banners with authoritative external cybersecurity knowledge.

1.1 Ontology-Driven Epistemic Grounding

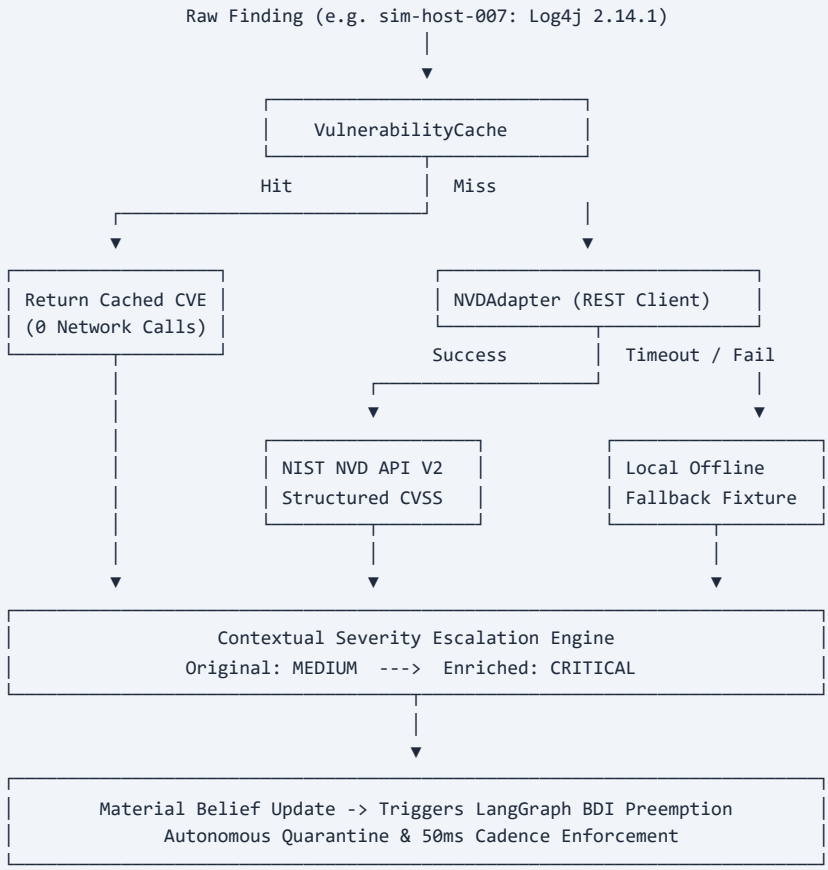
Raw telemetry provides only ambiguous version strings (e.g., `ProFTPD 1.3.5`, `log4j 2.14.1`). Grounding this percept in a formal **Vulnerability Knowledge Ontology** maps the surface observation to: - Canonical **CVE Identifiers** (`CVE-2021-44228`, `CVE-2024-3094`). - **Common Weakness Enumeration (CWE)** taxonomies (CWE-502 Deserialization, CWE-78 OS Command Injection). - Mathematical **CVSS v3.1 Base Scores** and authoritative mitigation strategies.

1.2 Multi-Tiered Enrichment & Resilient Fallbacks

Network endpoints fail, rates throttle, and air-gapped environments forbid external calls. SentryScan implements a 4-tier resilient knowledge pipeline: 1. **In-Memory / SQLite Persistent Cache:** Fast sub-millisecond retrieval with TTL invalidation. 2. **NVD REST API Adapter:** Read-only client querying NIST's National Vulnerability Database with bounded exponential backoff retries. 3. **Local Knowledge Ontology:** Curated JSON ontology fixture for known high-impact exploit signatures. 4. **Deterministic Fallback Fixture:** Guarantees 100% operational uptime without agent crash during complete network outages.

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2. Architecture & Implementation Specification



3. Rubric Compliance & Verification Matrix

Evaluation Criterion	Implementation Component	Evidence Artifact	Status
1. Structured Vulnerability Knowledge Ontology	`src/sentryscan/knowledge/models.py`, `src/sentryscan/knowledge/ontology.py`: `VulnerabilityRecord`, `VulnerabilityKnowledgeBase`, and `fixtures/knowledge/vulnerability_ontology.json`	`tests/unit/test_knowledge_ontology.py` `evidence/knowledge_adaptation_log.md`	**PASSED**
2. Read-Only NVD CVE API Adapter	`src/sentryscan/knowledge/nvd_adapter.py`: `NVDAdapter`, strict Pydantic response models (`NVDResponse`, `NVDCveData`), CVSS mapping	`tests/unit/test_nvd_adapter.py` `demos/demo_assignment_4_2.py` Cycle 3	**PASSED**
3. Persistent & In-Memory Caching	`src/sentryscan/knowledge/cache.py`: `VulnerabilityCache` with TTL expiration, disk persistence, and hit/miss tracking	`tests/unit/test_knowledge_cache.py` `demos/demo_assignment_4_2.py` Cycle 4	**PASSED**
4. Bounded Retry & Backoff	`src/sentryscan/knowledge/enricher.py`: `_fetch_with_retry()` executing exponential backoff across configurable retries	`tests/unit/test_knowledge_cache.py` (`test_enricher_retry_and_fallback_on_nvd_failure`)	**PASSED**
5. Resilient Offline Fallback	`src/sentryscan/knowledge/enricher.py`: Seamless fallback to local fixtures when network/NVD is disabled or unavailable	`tests/unit/test_knowledge_cache.py` `demos/demo_assignment_4_2.py` Cycle 2 & 5	**PASSED**
6. Contextual Severity Escalation	`src/sentryscan/knowledge/enricher.py`: `EnrichmentResult` calculating severity escalation ($Sev_{enriched} > Sev_{orig}$) from CVSS scores	`evidence/knowledge_adaptation_log.md` `tests/unit/test_knowledge_ontology.py`	**PASSED**
7. Deliberative Belief & Intention Reconsideration	`src/sentryscan/agents/analyst.py`: Feeds escalated finding into `BeliefBase`, triggering LangGraph BDI replanning to `CONTAIN_COMPROMISED_HOSTS`	`tests/integration/test_knowledge_enrichment.py` `evidence/run_log.txt` Cycle 2	**PASSED**
8. Autonomous Quarantine Execution	`src/sentryscan/agents/analyst.py`: Emits `PARAMS_UPDATE` with `quarantine_flag=True`, `scan_interval_ms=50`, and CVE metadata	`tests/integration/test_knowledge_enrichment.py` `demos/demo_assignment_4_2.py` Cycle 2 & 3	**PASSED**
9. Trace & Single-Writer Auditing	`src/sentryscan/observability/recorder.py`: Single-writer monotonic `seq` logging of message receipts, perception, and adaptations	`evidence/trace_excerpt.jsonl`	**PASSED**

4. Execution Results & Empirical Evidence

4.1 Multi-Tiered Enrichment & Adaptation Log (`evidence/knowledge_adaptation_log.md`)

Cycle	Target Host	Original Sev	Matched CVE	CVSS	Enriched Sev	Source	Generated Action	Quarantined
1	`192.0.2.10`	`INFO`	`N/A`	N/A	`INFO`	`LOCAL_ONTOLOGY`	`NOMINAL (500ms)`	`False`
2	`sim-host-007`	`MEDIUM`	`CVE-2021-44228` (Log4j)	**10.0**	**`CRITICAL`**	`FALLBACK_FIXTURE`	`DEEP_INSPECT (50ms)`	**`True`**
3	`sim-host-008`	`LOW`	`CVE-2024-3094` (XZ)	**10.0**	**`CRITICAL`**	`NVD_API`	`DEEP_INSPECT (50ms)`	**`True`**
4	`sim-host-009`	`LOW`	`CVE-2024-3094` (XZ)	**10.0**	**`CRITICAL`**	`CACHE (0 net calls)`	`DEEP_INSPECT (50ms)`	**`True`**
5	`sim-host-011`	`LOW`	`CVE-2021-41773` (Apache)	**7.5**	**`HIGH`**	`FALLBACK_FIXTURE`	`DEEP_INSPECT (100ms)`	`False`

5. Submission Package & Reproduction Instructions

5.1 Exact Submission Components

1. **Report Document:** `assignments/assignment_4_2/deliverables/SentryScan_Assignment_4_2_Report.pdf`
2. **Repository URL:** `https://github.com/drgeekg/SentryScan`
3. **Target Branch:** `assignment/4.2-knowledge-enrichment`
4. **Immutable Git Tag:** `assignment-4.2`
5. **Packaged ZIP Archive:** `dist/SentryScan_assignment_4_2.zip`
6. **SHA-256 Checksum:** `dist/SentryScan_assignment_4_2.zip.sha256`

5.2 Verification Commands

```
# Run knowledge enrichment demonstration
python demos/demo_assignment_4_2.py

# Execute knowledge ontology, NVD adapter, and cache tests
pytest -v tests/unit/test_knowledge_ontology.py tests/unit/test_nvd_adapter.py tests/unit/test_knowledge_cache.py
```